Continual Learning with O-LoRA: Orthogonal Low-Rank Adapters for Sequential Tasks

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1. Introduction

Continual learning requires a model to sequentially acquire knowledge across multiple tasks without forgetting what it previously learned. A common challenge is **catastrophic forgetting**: adapting to a new task overwrites representations built for earlier ones.

When parameter-efficient methods such as **LoRA** are applied in this setting, the low-rank adapter matrices trained on successive tasks tend to occupy overlapping subspaces, causing destructive interference.

O-LoRA addresses this by imposing an orthogonality constraint on the A matrices of each new task adapter with respect to all previously learned ones, ensuring each task projects inputs into a distinct region of the low-rank latent space and limiting the interference that drives forgetting.

2. Dataset and Preprocessing

Task sequence (fixed order):

- **AG News** – 4-class news topic classification
- **Amazon Polarity** – binary sentiment classification
- **DBpedia 14** – 14-class Wikipedia topic classification
- **Yahoo Answers Topics** – 10-class question topics

Each task is trained on independently, with no replay of prior task data.

Instruction-tuning prompt format:

```
{task instruction}\n
Option: {comma-separated label names}\n
{input text}\n
Answer:
```

Only the input text is truncated when the sequence exceeds the maximum length of 256. Cross-entropy loss is computed only on the label token; at evaluation the predicted label is the candidate label token with maximum logit at the position before `Answer`.

3. Models

Backbone: Qwen2.5-1.5B. **LoRA:** rank r=8, α=32, dropout 0.1, on Q, V of every attention layer. **Optim:** 1 epoch/task, lr 10^{-3}, batch 8.

Compared methods:

- **IncLoRA** – separate frozen adapter per task (BWT = 0 by construction).
- **O-LoRA** – one cumulative adapter; orthogonality penalty between the new task's A and all previous ones, weighted by $\lambda_1 \in [0.2, 0.3]$.

Orthogonal loss. For task t with $\Delta W_t = B_t A_t$:

$$\mathcal{L}_t = \mathcal{L}_{\mathrm{CE}}(A_t, B_t) + \lambda_1 \sum_{i<t} \left\| A_t A_i^{\top} \right\|_F^2$$

The penalty drives row-spaces of successive A into orthogonal directions.

4. Results

IncLoRA baseline (no forgetting):

Task	Accuracy
AG News	0.8945
Amazon Polarity	0.958
DBpedia 14	0.968
Yahoo Answers Topics	0.7135

Impact of prompt structure:

Prompt format	Avg Acc	BWT
Old (no label options)	0.597	−0.344
New (instruction + options)	0.799	−0.091

O-LoRA – summary metrics:

Samples / task	λ_1	Avg Acc	BWT
500/500/500/500	0.3	0.846	−0.032
500/650/1850/1300	0.2	0.799	−0.091
2100/2700/7500/5300	0.3	0.654	−0.268

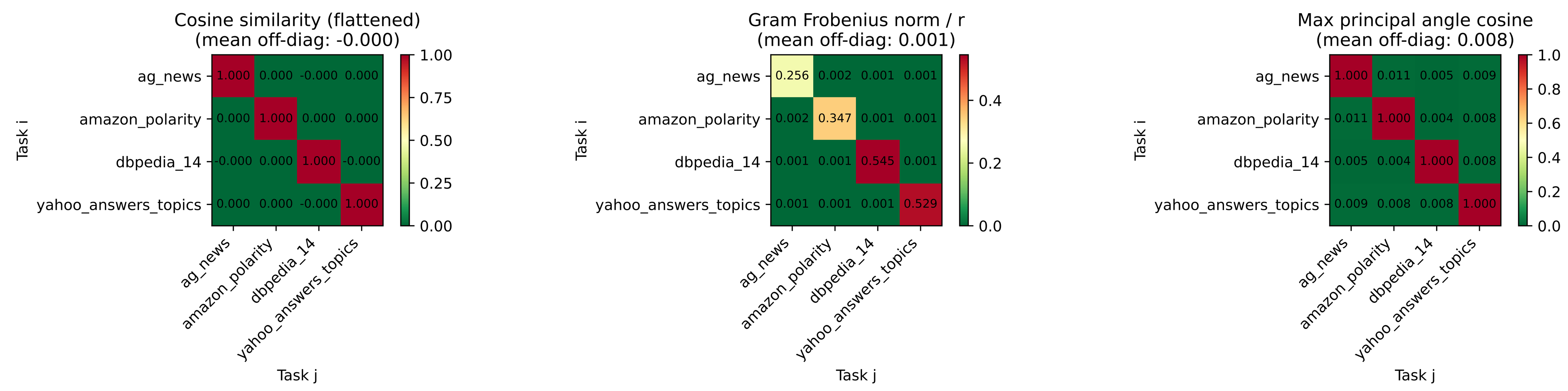
Per-task accuracy: initial → final

Task	Run 1	Run 2	Run 3
AG News	0.878→0.825 (−6%)	0.842→0.747 (−11%)	0.887→0.494 (−44%)
Amazon Polarity	0.956→0.957 (∼0%)	0.947→0.915 (−3%)	0.905→0.746 (−18%)
DBpedia 14	0.964→0.919 (−5%)	0.979→0.833 (−15%)	0.980→0.729 (−26%)
Yahoo Answers Topics	0.683→0.683 (—)	0.702→0.702 (—)	0.649→0.649 (—)

Avg accuracy and BWT both **degrade as the number of training samples grows**, even though more data improves *initial* per-task accuracy. The **instruction + options** prompt format yields +0.20 Avg Acc and +0.25 BWT – a larger effect than any hyperparameter choice.

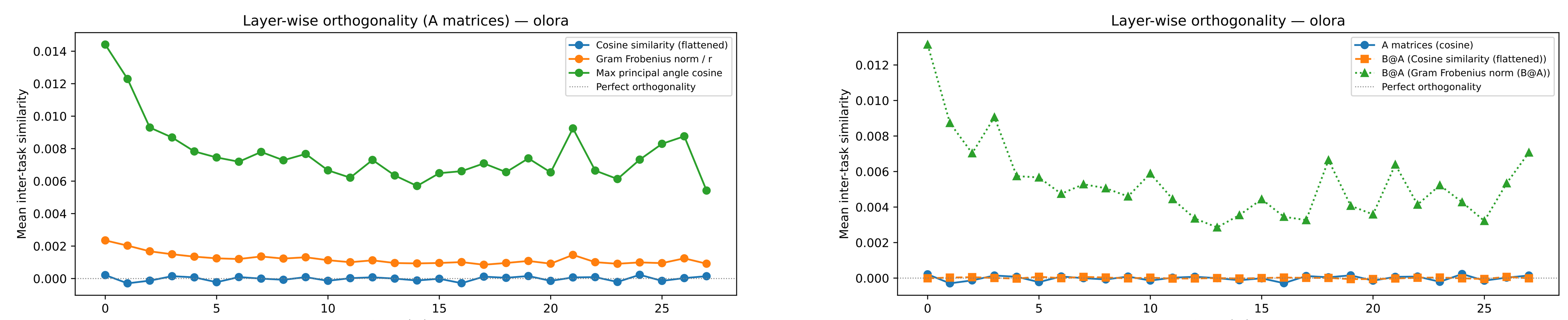
5. Orthogonality Analysis

Three pairwise metrics across task adapter matrices: cosine similarity, Gram Frobenius norm $/r$ (the quantity minimised by O-LoRA), and max principal angle cosine.



Task-pair similarity of A matrices (8h): **all metrics near-zero off-diagonal** – A matrices are geometrically separated.

Layer-wise breakdown (A vs BA). A different view – per-layer mean inter-task similarity. The constraint holds across all transformer blocks for A (left). The same cosine and Gram Frobenius metrics are also applied to the full weight update $\Delta W = BA$. For these product matrices the Frobenius norm is reported *without* rank normalisation – would scale values by up to 192× depending on the module.



B matrices are unconstrained: they share ∼ 30% of their column-space directions (principal-angle cosine ∼ 0.29) and grow in magnitude with training.

6. Conclusion

- A matrices are **geometrically separated** with mild penalties, uniformly across every layer.
- Extending the constraint to all six projection modules (q, k, v, o, up, down): A keeps a **similar degree of inter-task orthogonality regardless of the targeted module**.
- Forgetting is **highly sensitive to data scale**: BWT −0.032 at 500 samples/task, −0.268 at ∼50% of the paper's scale. **Prompt structure > hyperparameters**: +0.20 Avg Acc, +0.25 BWT. **Future work**: regularise B or constrain $\|BA\|$ directly.